em_q3

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Load function code

```
function_path <- file.path("R", "functions_for_em.R")

source(function_path)
```

Create data

```
delta_mat <- matrix(c(0,0,0,
                      1,0,0,
                      0,1,0,
                      0,0,1,
                      1,1,1), nrow = 5, byrow = T)

# Frequency for agreement patterns
G <- c(172738,2030,114,68,50)
```

Run EM

```
set.seed(123)

fit <- em_algorithm(
  delta_mat = delta_mat,
  G = G,
  n_starts = 50,
  tol = 1e-10,
  max_iter = 10000
)

save(fit, file = "results/em_fit.RData")
```

Report parameter estimates

```r
load("results/em_fit.RData")

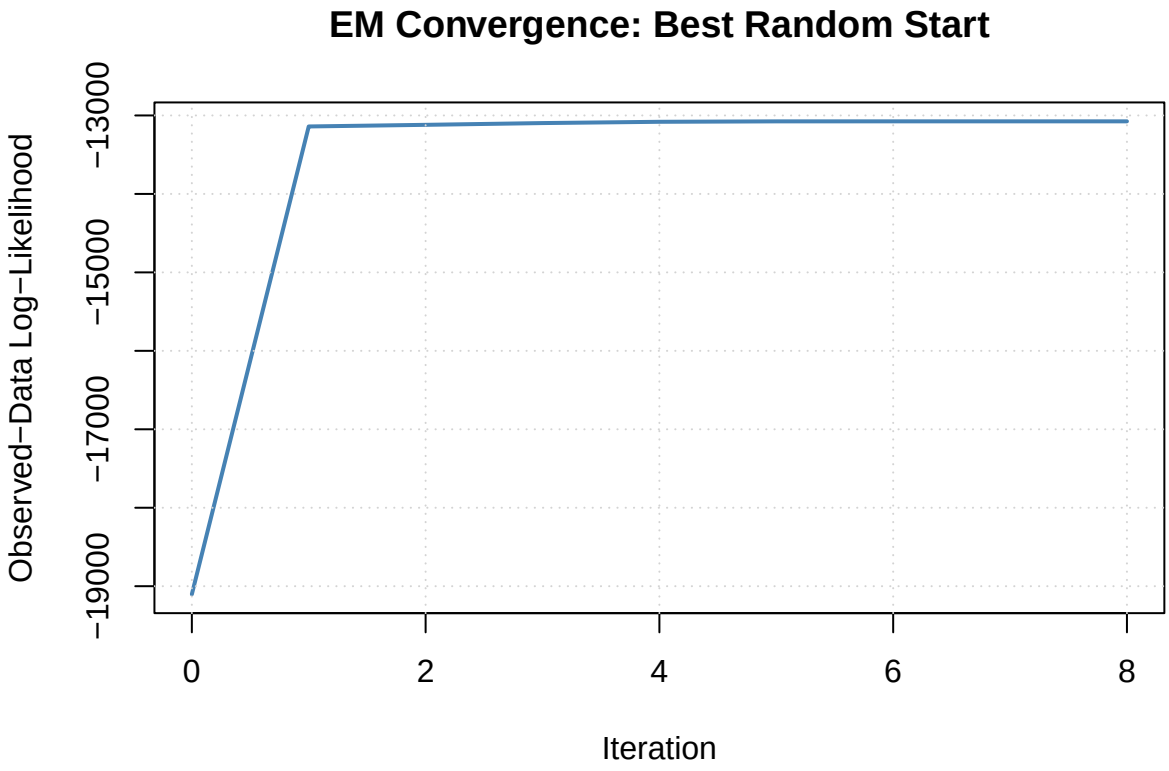
data.frame(
  delta = c(1,2,3),
  p_est = fit$p,
  m_est = fit$m,
  u_est = fit$u
)
```

```
##   delta        p_est m_est        u_est
## 1     1 0.0002857113     1 0.0116033181
## 2     2 0.0002857113     1 0.0006516177
## 3     3 0.0002857113     1 0.0003886854
```

Plot observed-data log-likelihood convergence

```r
plot(
  0:(length(fit$loglik) - 1),
  fit$loglik,
  type = "l",
  col = "steelblue",
  lwd = 2,
  xlab = "Iteration",
  ylab = "Observed-Data Log-Likelihood",
  main = "EM Convergence: Best Random Start"
)

grid()
```


EM Convergence: Best Random Start
Observed−Data Log−Likelihood
−13000
−15000
−17000
−19000
0
2
4
6
8
Iteration